

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE				
Regular/Supplementary Winter Examination – 2024				
Course: B. Tech		Branch: Electronics and Computer Engineering/ Electronics and Computer Science		
Semester :3 rd				
Subject Code & Name: BTESC305 & Digital Electronics and Microprocessor				
Max Marks: 60		Date:21/02/2025		Duration: 3 Hr.
Instructions to the Students: 1. Each question carries 12 marks. 2. Question No. 1 will be compulsory and include objective type questions. 3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6. 4. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question. 5. Use of non-programmable scientific calculators is allowed. 6. Assume suitable data wherever necessary and mention it clearly.				
			(Level/CO)	Marks
Q. 1	Objective type questions. (Compulsory Question)			12
1	Which of the following is a type of digital logic circuit?			1
	a. Combinational logic circuits	b. Sequential logic circuits	c. Both Combinational & Sequential logic circuits	d. None of these
2	Any signed negative binary number is recognized by its			1
	a. MSB	b. LSB	c. Nibble	d. Byte
3	Convert the hexadecimal number (1E2) ₁₆ to decimal.			1
	a. 482	b. 484	c. 481	d. 483
4	Which of the following is a universal gate			1
	a. AND	b. OR	c. NOT	d. NAND
5	What is the group of 1s in 4 cells of a K – map called?			1
	a. Pair	b. Quad	c. Octoate	d. None of these
6	Ripple counters are also called			1
	a. VLSI counters	b. SSI counters	c. Asynchronous counters	d. Synchronous counters
7	What kind of operation occurs in a J – K flip flop when both inputs J and K are equal to 1?			1
	a. Toggle operation	b. Preset operation	c. Reset operation	d. Clear operation
8	Which of the following is a non-vectored input?			1
	a. INTR	b. TRAP	c. RST 7.5	d. RST 6.5
9	How many flags are there in an 8085 microprocessor?			1
	a. 4	b. 5	c. 6	d. 7

10	Byte contains __ number of bits					1
	a. 4	b. 8	c.16	d. 32		
11	Which of the following is not a valid register pair in 8085 microprocessors?					1
	a. BC	b. DE	c. HL	d. AB		
12	8086 microprocessor is a ____ bit processor					1
	a. 4	b. 8	c. 16	d. 32		
Q. 2 Solve the following.						12
A) Draw the symbol & write truth tables of the following logic gates (i)NAND (ii) NOR (iii) EX-OR						6
B) State & Prove De-Morgan's theorem						6
Q.3 Solve the following.						12
A) Compare Multiplexer & Demultiplexer						6
B) Minimize & implement using K-Map $F(A, B, C, D) = \sum m(1,2,6,8,10,15) + d(0,3,5,11)$						6
Q. 4 Solve Any Two of the following.						12
A) Draw a neat diagram of SR Flip-flop & explain its operation						6
B) Design a 3-bit UP counter(ripple) using JK flip-flop						6
C) What is a shift register? What are its types & enlist its applications						6
Q.5 Solve Any Two of the following.						12
A) Draw the timing diagram of OP CODE fetch machine cycle						6
B) Perform the operation 7FH + 01H & indicate the status of the flags (8085 microprocessor)						6
C) Explain all addressing modes of 8085 microprocessor with examples						6
Q. 6 Solve Any Two of the following.						12
A) Draw a neat architecture of 8086 microprocessor & enlist its salient features						6
B) Explain flag register of 8086 microprocessor						6
C) Explain any two assembler directives						6

*** End ***